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Day 7: Probability - Assessment 1

Basic Probability:

Coin Toss:

Total outcomes = ~~S={H,T}~~ = S = {H, T} = 2

Favourable outcomes = {H} = 1

∴ Probability = $\frac{1}{2}$ = 0.5 for getting Head.
= (ans)

Dice Roll:

Total outcomes = S = {1, 2, 3, 4, 5, 6} = 6.

Favourable ~~os~~ x = {x | x > 4} = {5, 6} = 2.

∴ Probability = $\frac{2}{6}$ = $\frac{1}{3}$ (ans)

Conditional Probability:

Two Tosses:

Outcomes where 2nd toss is tail = {HT, TT} = 2

favourable outcome = {HT} = 1

Probability of getting Heads first given that second toss is tails = $\frac{1}{2}$ = 0.5 (ans)

Card Draw

Total Red Cards = 13 Hearts + 13 Diamonds = 26.

Favourable Red cards = 13.

Probability that drawn red card is Heart = $\frac{13}{26}$ = $\frac{1}{2}$ (ans)

Die Roll:

Total even number on die = {2, 4, 6} = 3

Favourable outcome = {2} = 1

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$\therefore$ Probability that the rolled even number is $2 = \frac{1}{3}$ (ans)

Bay's Theorem:

Coin Bag Example:

$P(Bag\ 1) = P(Bag\ 2) = \frac{1}{2}$ [Probability that coin is from Bag 1] $= P(Bag\ 1)$
[$P(Bag 2)$ = coin is from Bag 2].

$P(H|Bag\ 1)$ = Heads given that coin from Bag 1
$= \frac{1}{2}$.

$P(H|Bag\ 2)$ = Heads given that coin from Bag 2.
$= 1$.

$\therefore$ $P(Bag2|H)$ = Coin from Bag 2 given that it is Head.

$$= \frac{P(H|B2) \cdot P(B2)}{P(H|B1) \cdot P(B1) + P(H|B2) \cdot P(B2)}$$

$$= \frac{1 \cdot 1/2}{1/2 \cdot 1/2 + 1 \cdot 1/2} = \frac{2}{3} \text{ (ans)}$$

Medical Test:

$P(\text{Diesease}) = \frac{1}{1000}$ $P(\text{No diesease}) = \frac{999}{1000}$

$P(Pos|\text{Diesease}) = 0.99$ $P(Pos|\text{No Diesease}) = 0.01$
[Pos = Positive Test Result].

$$\therefore P(\text{Diesease}|Pos) = \frac{0.99 \cdot \frac{1}{1000}}{0.99 \cdot \frac{1}{1000} + 0.01 \cdot \frac{999}{1000}} \approx 0.09 = 9\% \text{ ans}$$

Weather Prediction:

P(Rain) = 0.7.

P(Correct | Rain) = 0.9

P(Wrong | No rain) = 0.2.

⇒ P(Correct | No rain) = 0.8.

P(Rain and correct) = 0.7 × 0.9 = 0.63

$P(\text{Correct} \mid \text{Rain}) = \frac{0.63}{0.7} = 0.9 = \underline{90\%}$ ans.

Student Test Result:

P(Prepared) = 0.7

P(Unprepared) = 0.3

P(Pass | Prepared) = 0.9 ; P(Pass | Unprepared) = 0.3

$P(\text{Prepared} \mid \text{Pass}) = \frac{0.9 \cdot 0.7}{0.9 \cdot 0.7 + 0.3 \cdot 0.3} = \frac{0.63}{0.77}$

$= 0.875 = 87.5\%$ ans

Bonus: Conditional Probability from Joint Probability

P(Boy) = 0.6   P(Girl) = 0.4.

P(Math | Boy) = 0.7   P(Math | Girl) = 0.5.

Total P(Math) = 0.6×0.7 + 0.4×0.5 = 0.42 + 0.2
= 0.62.

$\therefore P(\text{Boy} \mid \text{Math}) = \frac{0.6 \times 0.7}{0.62} \approx 0.677 = 67.7\%$ ans.